

SolveSteps()

simulation finished,
or solver failed or stopped?

yes

return True if the end time
was reached, otherwise False

no

state.startOfStep
= state.current

step finished?

yes

increment the
step counter

no

update the
current time
InitializeStep()

DiscontinuousIteration()?

failed

reduce the step (return False
at the minimum step size)
state.current = state.startOfStep

ok

FinishStep(); check whether
the step size may grow

